

Scout Agent

Project overview

This project involves building an automated monitoring agent using LangChain (or LangGraph) to assist agency recruiters. Recruiters interact individually with their personal "Scout Agent" via WhatsApp, instructing it to periodically check specified web pages (such as job postings, client websites, or competitor pages) at recruiter-defined frequencies (e.g., daily or multiple times per day). When a recruiter submits a task request, the Scout Agent first paraphrases and confirms the assignment clearly through WhatsApp. Upon confirmation, the assignment—including the original prompt, recruiter ID, target URL, task ID, status, and scheduling frequency—will be stored in a structured SQL database hosted on Azure.

The system will periodically execute tasks according to their scheduled frequency. It retrieves the latest content from each assigned URL, compares it with previously stored page snapshots, and performs an on-the-fly delta comparison via LangChain or an equivalent model. If a significant difference is detected based on the original task prompt criteria, the agent alerts the recruiter through a structured WhatsApp message template (using the existing 360dialog WhatsApp API integration). This initial notification offers options such as "Yes," "No," or "Maybe Later" to review the detected changes. Upon confirmation by the recruiter, detailed updates—including links—are communicated directly via WhatsApp. Historical snapshots of web page content, along with timestamps, are stored in the database to facilitate accurate comparisons.

Why Alert-Based Features Matter for Us

This feature provides us with a valuable position outside of the ATS ecosystem, avoiding overlap and making integration unnecessary. It allows us to maintain an ongoing communication loop with recruiters via WhatsApp, even if they don't actively use our browser extension. Once alerts are set, we deliver recurring value through automated updates, keeping us top-of-mind and embedded in their workflow. It creates a persistent communication channel where we can later introduce feature updates or additional capabilities without requiring a UI or setup. It requires no installation, no ATS integration, and raises no GDPR concerns, making it easy to activate. This also strengthens our outreach in LinkedIn campaigns by offering immediate utility with minimal friction. Over time, this recurring value builds trust and opens up opportunities for product expansion, referrals, and brand visibility in the recruiter's daily environment.

User story 1: Vacancy Monitoring and Alerts

User Story:

As an **agency recruiter** specialized in local government roles in the Netherlands,

I want to set up specific monitoring tasks in my **Scout Agent**, defining exactly **which local government vacancy pages** must be checked, **including how frequently (e.g., every hour, three times a day)**.

The Scout Agent should confirm these monitoring tasks clearly via WhatsApp, store each task, URL, and checking frequency in our Azure SQL database, and precisely execute the checks as scheduled.

Goal & Benefit (Why):

When a local government (e.g., gemeente Amsterdam, Rotterdam, or Utrecht) posts a new vacancy, I want to receive a timely WhatsApp notification from the Scout Agent, including the direct link. This enables me to immediately reach out to hiring managers, source candidates, and prepare candidate proposals—giving me a significant competitive advantage by responding faster than other recruiters.

Example Websites for Monitoring:

- <https://werkenbij.amsterdam.nl/vacatures>
- <https://www.werkenvoorrotterdam.nl/vacatures>
- OR
- <https://werkenbij.amsterdam.nl/vacatures?pageNumber=1&Keywords=&vakgebied=10>
- <https://www.werkenbijutrecht.nl/vacatures>

Acceptance Criteria:

- Recruiter defines the monitoring task clearly via WhatsApp, including frequency (e.g., "Check every 2 hours" Check every day once at 8:00").
- Scout Agent confirms task and frequency clearly via WhatsApp, stores these precisely in the Azure SQL database.
- Scout Agent retrieves content precisely at the defined frequency.
- Scout Agent accurately compares retrieved content with previously stored snapshots and immediately identifies new vacancies.
- Recruiter is notified instantly via WhatsApp if a new vacancy appears, clearly including a link to the new posting.

Sample scenario 1: Vacancy Monitoring and Alerts

Recruiter:

Hi Scout, please check <https://www.werkenvoorrotterdam.nl/vacatures> every 2 hours.

Scout Agent:

You want me to monitor:

<https://www.werkenvoorrotterdam.nl/vacatures>

Task: Notify you if a new job appears.

Check frequency: Every 2 hours

Shall I go ahead and set this up?

Buttons:

[ Confirm] [ Edit] [ Cancel]

Recruiter clicks: Confirm

Scout Agent:

Task created. I will check the page every 2 hours and alert you when new vacancies are posted.

(Later that day)

User story 2: Client News Tracker (Google News Monitoring)

User Story:

As an **agency recruiter** managing multiple key clients, I want my **Scout Agent** to monitor Google News for significant updates related specifically to my important client companies (e.g., ASML, ABN Amrobank), so that I receive timely, condensed WhatsApp notifications highlighting relevant news developments, including the ability to easily request a more detailed summary if desired.

Goal & Benefit (Why):

Staying informed about key developments at my clients enables me to communicate more effectively and confidently with my contacts. Being proactively aware of significant news ensures stronger relationships, enhances credibility, and eliminates manual daily checks, saving valuable time.

Example Companies to Monitor:

- ASML
- ABN Amrobank
- Philips
- ING
- Shell

Acceptance Criteria:

- Recruiter clearly specifies via WhatsApp the companies/topics for news monitoring (e.g., "Monitor significant news about ASML and ABN Amrobank daily").
- Scout Agent confirms each monitoring assignment, clearly summarizing the client's request via WhatsApp, and stores these assignments precisely in the Azure SQL database.
- Scout Agent performs news checks at recruiter-specified frequencies, identifying relevant news articles using Google News.
- Scout Agent sends concise, structured WhatsApp notifications summarizing significant news articles, with a simple option (e.g., "Read more") for recruiters to obtain a more detailed summary.
- Recruiter can easily adjust or add new companies/topics for monitoring at any time via WhatsApp interaction with the agent.

Sample scenario 1: Client News Tracker (Google News Monitoring)

Recruiter:

Scout, please monitor Google News for ASML every day.

Scout Agent:

You want me to monitor Google News for:

 ASML

Task: Notify you if a new article appears.

Frequency: Daily

Shall I set this up?

Buttons:

[ Start Monitoring] [ Edit] [ Cancel]

Recruiter clicks:  Start Monitoring

Scout Agent:

 Monitoring task created for: ASML

I will check daily and notify you of significant updates.

User story 3: LinkedIn Contact Activity Monitor

User Story:

As an **agency recruiter** who maintains close relationships with important contacts (e.g., Directors, HR Managers, Hiring Managers),

I want my **Scout Agent** to regularly monitor the LinkedIn profiles of specific individuals I specify, so that I receive timely WhatsApp notifications whenever these contacts share new updates or posts.

Goal & Benefit (Why):

Being instantly aware of new LinkedIn activity from key contacts helps me to remain informed about their professional developments and interests. This allows me to engage in more relevant, timely, and personalized conversations, ultimately enhancing my relationship with these contacts and increasing my professional effectiveness without needing daily manual checks.

Example Profiles to Monitor:

- (Recruiter will specify URLs of LinkedIn profiles directly.)

Acceptance Criteria:

- Recruiter provides specific LinkedIn profile URLs via WhatsApp to monitor.
- Scout Agent confirms clearly via WhatsApp the monitoring task, summarizing exactly which profiles will be tracked, storing these assignments in the Azure SQL database.
- Scout Agent checks specified LinkedIn profiles at recruiter-defined intervals.
- Recruiter receives concise, structured WhatsApp notifications summarizing any new posts or significant activity from the monitored contacts.
- Notifications include a direct link to view the LinkedIn update, with an option for further details or summaries upon recruiter request.

Sample scenario 1: Client News Tracker (Google News Monitoring)

Recruiter:

Scout, I want to monitor updates from this LinkedIn profile:

<https://www.linkedin.com/in/esther-hr-director>

Scout Agent:

You want me to monitor LinkedIn for:



Esther – HR Director

Task: Notify you if a change on her profile appears.

How often should I check her profile?

Buttons:

[🕒 Daily at 08:00] [⌚ Every 4 Hours] [📝 Custom]

Recruiter clicks: 🕒 Daily at 08:00

Scout Agent:



Monitoring task created for Esther – HR Director

I'll check the profile daily at 08:00 and notify you of any new posts or updates.